SIMATS ENGINEERING

ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	**DHILIBAN M**
Register Number	**19242166**

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 50**

Question Count: 5

Code of Conduct

I ___________DHILIBAN M/192421166_________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ____________________

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation

		key terms			
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1.Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Cloud computing is the delivery of computing resources such as servers, storage, databases, networking, software, and analytics over the Internet on a pay-as-you-use basis. Instead of purchasing and maintaining physical hardware, users can access these resources whenever needed.

Characteristics:

1. On-Demand Self-Service

- Users can create or remove computing resources without contacting the service provider.

2. Broad Network Access

- Services are accessible through the Internet using computers, laptops, tablets, and smartphones.

3. Resource Pooling

- Cloud providers share computing resources among multiple customers using a multi-tenant architecture.

4. Rapid Elasticity

- Resources can be increased or decreased quickly according to workload demand.

Difference from Traditional Computing

- Traditional computing requires purchasing, installing, and maintaining hardware.
- Cloud computing provides scalable resources over the Internet with minimal upfront investment.

Example: A startup using virtual servers from AWS instead of purchasing physical servers.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

Answer:

Cloud computing evolved through several technological advancements.

Stage 1: Hardware Evolution

- Organizations owned dedicated physical servers.
- High cost and low resource utilization.

Stage 2: Virtualization

- Multiple virtual machines could run on one physical server.
- Improved hardware utilization and reduced costs.

Stage 3: Distributed Computing

- Multiple computers worked together to perform large tasks.
- Improved scalability and reliability.

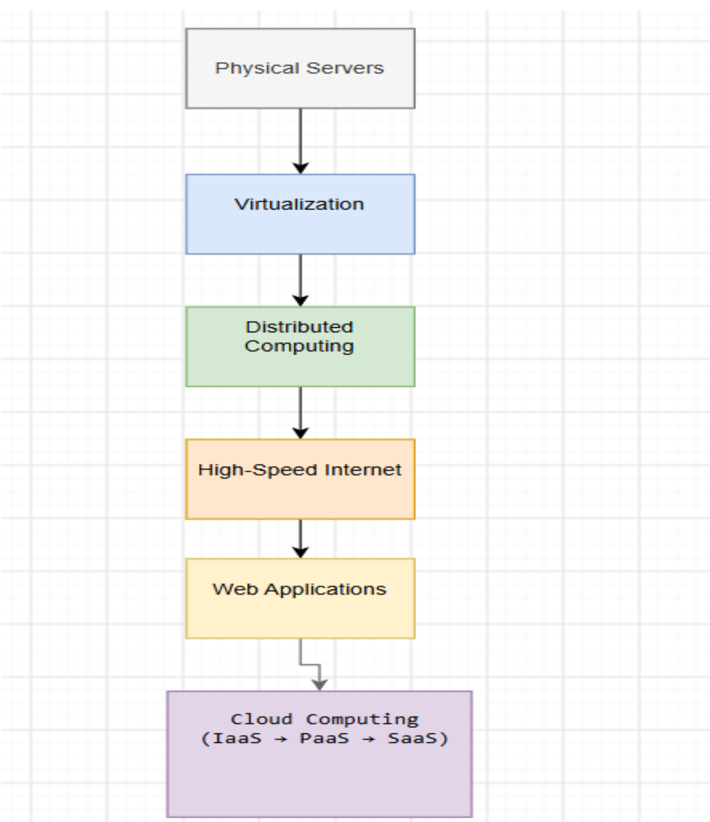
Stage 4: Internet Evolution

- High-speed Internet enabled remote access to computing resources.

Stage 5: Internet Software Evolution

- Applications became web-based and Software as a Service (SaaS) emerged.
- Cloud providers offered Infrastructure, Platform, and Software services.

These developments resulted in today's cloud platforms, which provide scalable, reliable, and cost-effective computing resources.



3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Answer:

Server virtualization is the process of dividing one physical server into multiple virtual machines (VMs) using a hypervisor. Each VM operates independently with its own operating system and applications.

Role in Cloud Computing

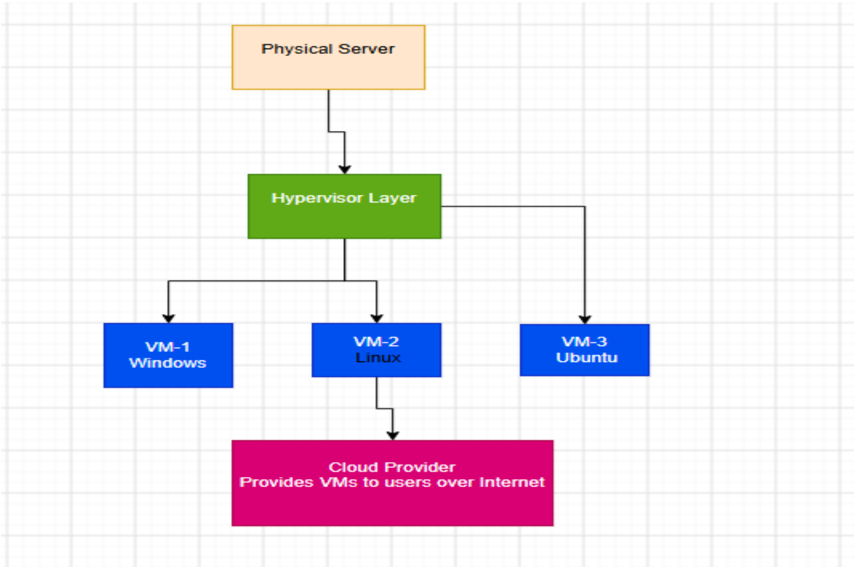
- Improves hardware utilization.
- Enables rapid deployment of virtual machines.
- Supports resource isolation.
- Reduces infrastructure costs.
- Makes cloud scalability possible.

Difference

Virtualization	**Cloud Computing**
Technology	Service delivery model
Creates multiple virtual machines	Delivers IT resources over the Internet
Uses hypervisors	Uses virtualization plus automation
Focuses on efficient hardware usage	Focuses on providing services to users
Can exist without cloud	Usually depends on virtualization

Example

- VMware creates virtual machines.
- AWS provides cloud services using virtualization.



4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.

Answer:

Feature	**SOAP**	**REST**
Full Form	Simple Object Access Protocol	Representational State Transfer
Communication	Protocol	Architectural Style
Data Format	XML only	JSON, XML, HTML
Speed	Slower	Faster
Complexity	High	Low
Security	WS-Security	HTTPS, OAuth
Flexibility	Less flexible	Highly flexible
Cloud Suitability	Enterprise applications	Cloud and mobile applications

Explanation

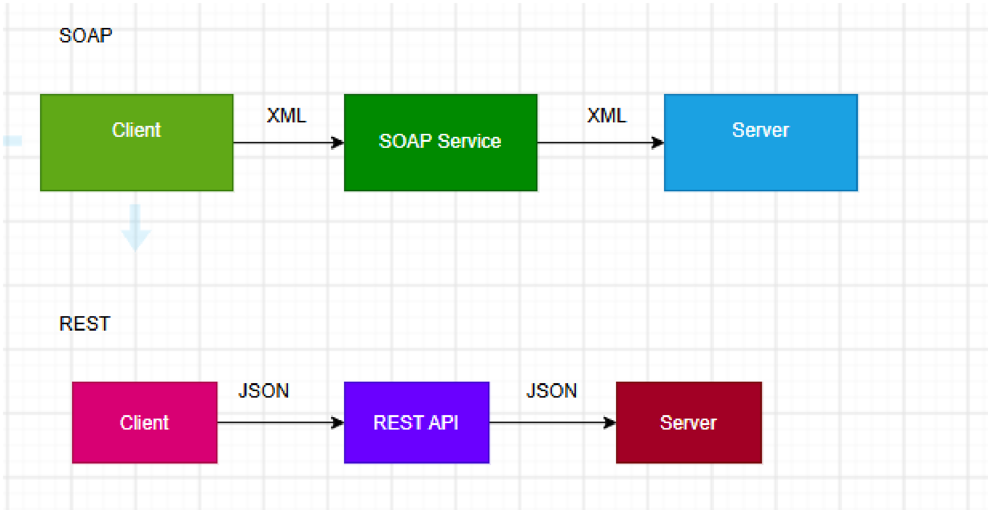
SOAP is a protocol designed for secure enterprise communication and uses XML messages.

REST is a lightweight architectural style that commonly uses JSON and is widely adopted in cloud computing because of its speed and simplicity.

Examples

- SOAP: Banking systems

- REST: Weather APIs, social media APIs



5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Answer:

Service Model	Description	Example
IaaS	Provides virtual machines, storage, and networking	Amazon EC2
PaaS	Provides a platform to develop and deploy applications	Google App Engine
SaaS	Provides ready-to-use software over the Internet	Google Docs
XaaS	Any IT service delivered over the Internet	Microsoft 365

Explanation

IaaS (Infrastructure as a Service)

- Users manage operating systems and applications.
- Cloud provider manages hardware.

PaaS (Platform as a Service)

- Developers focus on coding.
- Platform manages servers and runtime.

SaaS (Software as a Service)

- Users access software through a web browser.
- No installation required.

XaaS (Anything as a Service)

- Includes Storage as a Service, Database as a Service, Security as a Service, and many others.